Class XII Maths Sample Paper – Set 2

Time Allowed: 30 Minutes

General Instructions:

1. This Question paper contains - five sections A, Each section is compulsory. However, there are internal choices in some questions.

2. **Section A** has 18 MCQ's and 02 Assertion-Reason based questions of 1 mark each.

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Section A		20
Q. No.	**Questions**	**Marks**
1.	If A = [a_{ij}] is a scalar matrix of order n × n such that a_{ij} = k for all i, then trace of A is equal to a) $\begin{matrix} n \\ k \end{matrix}$ b) n – k c) nk d) n + k	1
2.	Three points P(2x, x + 3), Q(0, x) and R(x + 3, x + 6) are collinear, then x is equal to: a) 2 b) 0 c) 3 d) 1	1
3.	If A is a 3 × 3 matrix and |A| = -2, then value of |A (adj A)| is a) -2 b) 8	1

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	c) 2 d) -8	
4.	If A is a non-singular matrix and A' denotes the transpose of A, then a) $\lvert AA' \rvert \neq \lvert A^2 \rvert$ b) $\lvert A \rvert - \lvert A' \rvert \neq 0$ c) $\lvert A \rvert + \lvert A' \rvert \neq 0$ d) $\lvert A \rvert \neq \lvert A' \rvert$	1
5.	The direction ratios of a line parallel to z-axis are: a) < 0, 0, 0 > b) < 1, 1, 0 > c) < 1, 1, 1 > d) < 0, 0, 1	1
6.	The general solution of the differential equation is $\frac{dy}{dx} = \frac{y}{x}$ a) y = kx b) log y - kx c) cos x d) tan x	1

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7.	The corner points of the feasible region determined by the system of linear constraints are (0, 10), (5, 5), (15, 15), (0, 20). Let Z = px + qy, where p, q > 0. Condition on p and q so that the maximum of Z occurs at both the points (15, 15) and (0, 20) is a) q = 3p b) q = 2p c) p = q d) p = 2q	1
8.	In $\Delta\ ABC$, $\overrightarrow{AB} = \hat{i} + \hat{j} + 2\hat{k}$ and $\overrightarrow{AC} = 3\hat{i} - \hat{j} + 4\hat{k}$. If D is the mid-point of BC, then vector $\overrightarrow{AD}$ is equal to: a) $\hat{i} - \hat{j} + \hat{k}$ b) $2\hat{i} - 2\hat{j} + 2\hat{k}$ c) $4\hat{i} + 6\hat{k}$ d) $2\hat{i} + 3\hat{k}$	1
9.	Evaluate the integral: $\int_0^{\frac{\pi}{4}} \frac{e^{\tan x}}{\cos^2 x} dx = ?$ a) $(e - 1)$ b) $\left(\frac{1}{e} + 1\right)$	1

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	c) $(e+1)$ d) $\left(\frac{1}{e}-1\right)$	
10.	If $A=\begin{bmatrix}5 & x\\ y & 0\end{bmatrix}$ and $A=A^T$, where A^T is the transpose of the matrix A, then: a) $x=0, y=5$ b) $x=5, y=0$ c) $x=y$ d) $x+y=5$	1
11.	The solution set of the inequality 3x + 5y < 4 is a) an open half-plane not containing the origin. b) an open half-plane containing the origin. c) a closed half plane containing the origin. d) the whole XY-plane not containing the line 3x+ 5y = 4	1
12.	If the projection of $\vec{a}=\hat{i}-2\hat{j}+3\hat{k}$ on $\vec{b}=2\hat{i}+\lambda\hat{k}$ is zero, then the value of λ is: a) 0 b) 1	1

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	c) $-\frac{3}{2}$ d) $-\frac{2}{3}$	
13.	Evaluate the determinant: $\begin{vmatrix} a+ib & c+id \\ -c+id & a-ib \end{vmatrix} = ?$ a) $(a^2-b^2-c^2-d^2)$ b) $(a^2+b^2+c^2-d^2)$ c) $(a^2-b^2+c^2-d^2)$ d) $(a^2+b^2+c^2+d^2)$	1
14.	X and Y are independent events such that $P(X \cap Y) = \frac{2}{5}$ and $P(X) = \frac{3}{5}$. Then $P(Y)$ is equal to: a) $\frac{2}{3}$ b) $\frac{1}{3}$ c) $\frac{1}{5}$ d) $\frac{2}{5}$	1

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15.	The solution of the differential equation $\frac{dy}{dx} = e^{x-y} + x^2 e^{-y}$ is: a) $e^{x} + e^{y} = \frac{x^3}{3} + c$ b) $e^{x} - e^{y} = \frac{x^3}{3} + c$ c) $y = e^{x-y} - x^2 e^{y} + c$ d) $e^{y} - e^{x} = \frac{x^3}{3} + C$	1
16.	If $\lvert \vec{a} \rvert = 4$ and $-3 \le \lambda \le 2$, then the range of $\lvert \lambda \vec{a} \rvert$ is: a) [0, 12] b) [0, 8] c) [8, 12] d) [-12, 8]	1
17.	If $f(x) = \begin{cases} kx + 5, & \text{when } x \le 2, \\ x + 1, & \text{when } x > 2, \end{cases}$ is continuous at $x = 2$, then $k = ?$ a) -2 b) -1 c) 2 d) -3	1

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18.	If the lines $\frac{x-1}{-3}=\frac{y-2}{2k}=\frac{z-3}{2}$ and $\frac{x-1}{3k}=\frac{y-1}{1}=\frac{z-6}{-5}$ are perpendicular to each other, then $k=?$ a) $-\frac{10}{7}$ b) $\frac{5}{7}$ c) $-\frac{5}{7}$ d) $\frac{10}{7}$	1
19.	**Assertion (A):** The rate of change of area of a circle with respect to its radius r when r = 6 cm is 12 cm^2/cm. **Reason (R):** Rate of change of area of a circle with respect to its radius r is $\frac{dA}{dr}$, where A is the area of the circle. a) Both A and R are true and R is the correct explanation of A. b) Both A and R are true but R is not the correct explanation of A. c) A is true but R is false. d) A is false but R is true	1
20.	**Assertion (A):** Let A ={2, 4, 6} and B = {3, 5, 7, 9} and defined a function f = {(2, 3), (4, 5), (6, 7)} from A to B. Then, f is not onto.	1

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	Reason (R): A function f : A → B is said to be onto, if every element of B is the image of some elements of A under f. a) Both A and R are true and R is the correct explanation of A. b) Both A and R are true but R is not the correct explanation of A. c) A is true but R is false. d) A is false but R is true	

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